

Noise, Mixed States, and Error Mitigation

EVA: Quantum Machine Learning – Lecture 13

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Spring 2026

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1. Why pure states are not enough on hardware
2. Density matrices: definition, measurement rule, examples
3. Quantum channels (noise models) via Kraus operators
4. Three canonical noise models: depolarizing, dephasing, amplitude damping
5. Consequences for QML pipelines
6. Error mitigation: measurement calibration, zero-noise extrapolation

Part 1. Why pure states are not enough on hardware

Simulators vs real hardware

In a statevector simulator, the state of n qubits is a unit vector $|\psi\rangle \in \mathbb{C}^{2^n}$, evolving as:

$$|\psi(t)\rangle = U(t) |\psi(0)\rangle .$$

This is exact and fully described by 2^n complex amplitudes.

On physical hardware, two qualitatively different error sources arise:

- **Coherent errors:** systematic, gate-specific rotations (e.g. a CNOT that over-rotates by a fixed angle). These are in principle correctable by calibration.
- **Incoherent errors (decoherence):** the qubit exchanges energy or phase information with its environment. The qubit is no longer in a definite pure state; it must be described by a mixture of pure states.

Incoherent errors require the density matrix formalism to describe correctly.

The environment causes decoherence

When qubit Q is entangled with environment E , the joint state is:

$$|\Psi\rangle_{QE} = \alpha |0\rangle_Q |e_0\rangle_E + \beta |1\rangle_Q |e_1\rangle_E.$$

If we have no access to E , we must trace over it to get the qubit's state:

$$\rho_Q = \text{Tr}_E(|\Psi\rangle_{QE} \langle\Psi|_{QE}) = |\alpha|^2 |0\rangle \langle 0| + |\beta|^2 |1\rangle \langle 1| + \alpha\beta^* \langle e_1|e_0|0\rangle \langle 1| + \dots$$

If the environment states $|e_0\rangle, |e_1\rangle$ are orthogonal (distinguishable records of which state the qubit was in), the off-diagonal terms vanish:

$$\rho_Q \rightarrow |\alpha|^2 |0\rangle \langle 0| + |\beta|^2 |1\rangle \langle 1|.$$

This is a **classical mixture**: the qubit appears to be in $|0\rangle$ with probability $|\alpha|^2$ and in $|1\rangle$ with probability $|\beta|^2$. All quantum coherence (the superposition) has been lost.

Part 2. Density matrices

Density matrix: definition

The density matrix is the general state descriptor for both pure and mixed quantum states.

Pure state:

$$\rho = |\psi\rangle \langle\psi|, \quad \rho^2 = \rho, \quad \text{Tr}(\rho) = 1.$$

Classical mixture (probabilistic uncertainty over pure states):

$$\rho = \sum_k p_k |\psi_k\rangle \langle\psi_k|, \quad p_k \geq 0, \quad \sum_k p_k = 1.$$

For a mixture: $\text{Tr}(\rho^2) < 1$ (equality only for pure states).

Properties that any density matrix must satisfy:

- Hermitian: $\rho = \rho^\dagger$
- unit trace: $\text{Tr}(\rho) = 1$
- positive semi-definite: $\langle\phi|\rho|\phi\rangle \geq 0$ for all $|\phi\rangle$

Worked 1-qubit example

Let $|\psi\rangle = R_y(\theta) |0\rangle = \cos(\theta/2) |0\rangle + \sin(\theta/2) |1\rangle$.

Pure state density matrix:

$$\rho = |\psi\rangle \langle\psi| = \begin{bmatrix} \cos^2(\theta/2) & \cos(\theta/2) \sin(\theta/2) \\ \cos(\theta/2) \sin(\theta/2) & \sin^2(\theta/2) \end{bmatrix}.$$

Expectation value via trace rule:

$$\langle Z \rangle = \text{Tr}(\rho Z) = \cos^2(\theta/2) - \sin^2(\theta/2) = \cos \theta.$$

Compare with the ket-bra formula: $\langle Z \rangle = \langle\psi| Z |\psi\rangle = \cos \theta$. ✓

The maximally mixed state $\rho = I/2$ has $\text{Tr}(\rho Z) = \text{Tr}(Z/2) = 0$: no observable is predictable from the maximally mixed state.

lecture_13_demo.ipynb, Demo 1: trace rule and density matrix diagonals.

Bloch sphere picture for density matrices

Any single-qubit density matrix can be written as:

$$\rho = \frac{I + \mathbf{r} \cdot \boldsymbol{\sigma}}{2}, \quad \mathbf{r} = (\text{Tr}(\rho X), \text{Tr}(\rho Y), \text{Tr}(\rho Z)) \in \mathbb{R}^3.$$

$\mathbf{r}$ is the **Bloch vector**:

- pure states: $|\mathbf{r}| = 1$ (surface of the Bloch sphere)
- mixed states: $|\mathbf{r}| < 1$ (interior of the Bloch sphere)
- maximally mixed state: $\mathbf{r} = 0$ (center)

Noise channels act as linear maps on the Bloch vector: depolarizing shrinks $\mathbf{r}$ uniformly toward the center, dephasing shrinks only the x - and y -components, amplitude damping shifts $\mathbf{r}$ toward the north pole $|0\rangle$.

Part 3. Quantum channels (noise models)

Quantum channels and the Kraus representation

A **quantum channel** (completely positive, trace-preserving map) describes any physical evolution of a density matrix, including unitary gates and noise:

$$\mathcal{E} : \rho \mapsto \mathcal{E}(\rho) = \sum_k E_k \rho E_k^\dagger,$$

where the Kraus operators $\{E_k\}$ satisfy:

$$\sum_k E_k^\dagger E_k = I.$$

Special cases:

- **Unitary evolution:** single Kraus operator $E_1 = U$, $\mathcal{E}(\rho) = U\rho U^\dagger$
- **Projection (measurement):** $E_0 = |0\rangle\langle 0|$, $E_1 = |1\rangle\langle 1|$
- **Depolarizing noise:** 4 Kraus operators (see next slide)

The Kraus representation is not unique (different sets can describe the same channel).

Depolarizing noise: explicit Kraus form

The one-qubit depolarizing channel with rate $p \in [0, 1]$:

$$\mathcal{E}_{\text{dep}}(\rho) = (1 - p)\rho + \frac{p}{3}(X\rho X + Y\rho Y + Z\rho Z).$$

Equivalently:

$$\mathcal{E}_{\text{dep}}(\rho) = (1 - p)\rho + p \frac{I}{2}.$$

Effect on Pauli expectations: for any traceless observable $O \in \{X, Y, Z\}$:

$$\langle O \rangle_{\mathcal{E}(\rho)} = (1 - p) \langle O \rangle_{\rho}.$$

The Bloch vector is shrunk uniformly by factor $(1 - p)$: $r \rightarrow (1 - p)r$.

lecture_13_demo.ipynb, Demo 2: depolarizing shrinkage of Bloch vector.

Dephasing and amplitude damping: qualitative

Channel	Physical mechanism	Bloch vector effect	Which expectation change
Dephasing (T_2)	Random relative phase	$r_x, r_y \rightarrow 0$	$\langle X \rangle, \langle Y \rangle$ shrink
Amplitude damping (T_1)	Energy relaxation to $ 0\rangle$	$r \rightarrow (0, 0, +1)$	$\langle Z \rangle$ drifts upward
Depolarizing	Random Pauli error	$r \rightarrow 0$	all Paulis shrink

Dephasing (decoherence time T_2) is typically the dominant error channel for superconducting qubits at low temperatures. Amplitude damping (T_1) is important for long gate sequences and idle qubits.

Both T_1 and T_2 are hardware-specific parameters. Modern IBM superconducting qubits: $T_1 \approx 100\text{--}300\ \mu\text{s}$, $T_2 \approx 50\text{--}200\ \mu\text{s}$. Gate times: 2-qubit gates take $\approx 200\text{--}500\ \text{ns}$.

Part 4. Consequences for QML pipelines

Noise changes QML features

In a QML pipeline, the measured features are expectation values $\langle O_k \rangle = \text{Tr}(\rho O_k)$. Under a depolarizing channel with rate p :

$$\langle O_k \rangle_{\text{noisy}} = (1 - p)^D \langle O_k \rangle_{\text{ideal}},$$

where D is the circuit depth (number of noisy gate layers).

Consequences:

- **Attenuation:** features shrink exponentially with circuit depth
- **Bias:** the feature distribution shifts toward zero (maximally mixed)
- **Variance increase:** shot noise and quantum noise compound
- **Label corruption:** if features are input to a classifier, all features are pushed toward a single point – classes overlap more

Practical implication: on NISQ hardware, keep circuits shallow. If depth must increase, apply error mitigation.

Noise and nonlinearity: a clarification

A common misconception: “noise makes quantum circuits nonlinear.”

Density matrices and channels are **linear** in ρ :

$$\mathcal{E}(a\rho_1 + b\rho_2) = a\mathcal{E}(\rho_1) + b\mathcal{E}(\rho_2),$$

and $\langle O \rangle = \text{Tr}(\rho O)$ is linear in ρ .

Nonlinearity in QML pipelines enters from:

- the *parameter dependence* $\theta \mapsto \langle O \rangle(\theta)$, which is nonlinear in θ (trigonometric functions of gate angles)
- classical post-processing: loss functions, softmax, sigmoid
- conditional updates or post-selection on measurement outcomes

Noise does not add a new source of nonlinearity for learning purposes; it mainly attenuates and biases the existing (quantum) nonlinear features.

Circuit depth budget on NISQ hardware

Given a target feature quality (signal-to-noise ratio), the maximum usable circuit depth is constrained by the gate error rate ϵ :

$$\text{feature SNR} \approx \frac{(1 - \epsilon)^D}{\sqrt{1/S}},$$

where D is depth and S is shot count.

Gate error ϵ	Target SNR	Max depth D	Shots S needed
0.001 (very good)	10	≈ 2300	100
0.01 (typical IBM)	10	≈ 230	100
0.01	10	≈ 130	1000
0.05 (poor)	10	≈ 46	100

With typical IBM two-qubit gate fidelities of 99–99.5% ($\epsilon \approx 0.005$ – 0.01), circuits beyond depth 100–200 show significant feature attenuation without mitigation.

Part 5. Error mitigation

Error mitigation vs error correction

	Error correction	Error mitigation
Mechanism	Encodes logical qubits in physical qubits	Post-processing of noisy results
Hardware overhead	Large (50–1000 physical qubits per logical)	None (runs on existing hardware)
When available	Fault-tolerant era (future)	Now (NISQ)
Noise suppression	Exponential (in code distance)	Polynomial or constant
Computational cost	In hardware (additional qubits, gates)	Classical post-processing

Error mitigation techniques reduce bias in expectation value estimates without full error correction. They do not reduce the variance from shot noise, and may increase sampling cost.

Measurement error mitigation

Readout errors: the qubit is in state $|0\rangle$ but the detector reports 1 (or vice versa). These are characterized by a confusion matrix M :

$$\hat{p}_{\text{meas}} \approx M p_{\text{ideal}}, \quad M = \begin{bmatrix} P(0|0) & P(0|1) \\ P(1|0) & P(1|1) \end{bmatrix}.$$

Calibration procedure:

1. Prepare $|0\rangle$, measure 1000 times $\rightarrow$ estimate $P(1|0)$ (bit-flip rate)
2. Prepare $|1\rangle$ (via X gate), measure 1000 times $\rightarrow$ estimate $P(0|1)$
3. Invert M to recover $p_{\text{ideal}} \approx M^{-1} \hat{p}_{\text{meas}}$

For n qubits, the full confusion matrix is $2^n \times 2^n$ (intractable for large n). Practical approximations: tensored per-qubit calibration (assumes independent errors), or local neighborhood calibration.

Zero-Noise Extrapolation (ZNE)

ZNE (Temme et al. 2017, Li and Benjamin 2017) estimates the noiseless expectation value $\langle O \rangle(0)$ by evaluating $\langle O \rangle(\lambda)$ at multiple noise levels $\lambda > 1$ and extrapolating.

Step 1: Noise amplification. Scale the noise by inserting redundant gate pairs. For a gate G , the sequence $G, G^\dagger, G$ has the same ideal action as G but twice the noise. Noise scales $\lambda = 1, 3, 5, \dots$ are accessible.

Step 2: Extrapolation. Fit a polynomial or exponential to $\langle O \rangle(\lambda)$ and evaluate at $\lambda = 0$:

$$\langle O \rangle_{\text{mit}} = \lim_{\lambda \rightarrow 0} \hat{f}(\lambda).$$

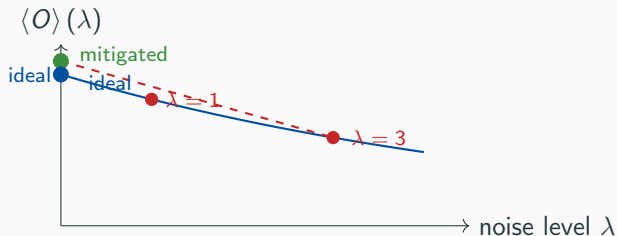
Extrapolation error depends on the noise model and the order of the fit. Richardson extrapolation with 3 points gives 4th-order accuracy in λ .

lecture_13_demo.ipynb, Demo 4: ZNE workflow on a toy observable.

ZNE in practice: linear extrapolation

For depolarizing noise with rate p per layer and D layers, $\langle O \rangle(\lambda) = (1 - \lambda p)^D \langle O \rangle_{\text{ideal}}$. Linear extrapolation at $\lambda = 1$ and $\lambda = 3$:

$$\langle O \rangle_{\text{mit}} = 2 \langle O \rangle(1) - \langle O \rangle(3) / 2 \dots$$



Beyond ZNE: probabilistic error cancellation and twirling

Probabilistic error cancellation (PEC): Expresses the ideal operation as a linear combination of noisy operations (including some with negative coefficients). Samples circuits according to these coefficients. Unbiased estimator of the ideal expectation value, but with cost $\mathcal{O}(e^{2\gamma})$ where γ is the “error budget.”

Pauli twirling / randomized benchmarking: Inserts random Pauli gates before and after each noisy two-qubit gate, conjugated so the ideal circuit is unchanged. This converts coherent errors into stochastic Pauli noise, which is easier to model and mitigate. Used as a preprocessing step before ZNE or PEC.

Qiskit (via the Qiskit IBM Runtime) implements ZNE, PEC, and twirling as options in the EstimatorV2 interface. These are enabled by passing a ResilienceOptions object.

Summary

Topic	Key takeaway
Pure vs mixed	Pure states break under decoherence; density matrix is the full description
Density matrix	Hermitian, trace-1, PSD; $\text{Tr}(\rho^2) = 1$ iff pure
Bloch vector	$\rho = (I + \mathbf{r} \cdot \boldsymbol{\sigma})/2$; noise shrinks $\mathbf{r}$
Kraus operators	$\mathcal{E}(\rho) = \sum_k E_k \rho E_k^\dagger$, $\sum_k E_k^\dagger E_k = I$
Depolarizing	Shrinks all Pauli expectations by $(1 - p)$; Bloch sphere contraction
Dephasing / T2	Destroys $\langle X \rangle$, $\langle Y \rangle$; preserves $\langle Z \rangle$
Amplitude damping / T1	Relaxes qubit toward $ 0\rangle$; shifts $\langle Z \rangle$ upward
QML impact	Exponential feature attenuation with depth; classes overlap
Measurement mitigation	Calibration matrix inversion to correct readout errors
ZNE	Noise scaling + extrapolation; reduces bias, not variance

1. Nielsen and Chuang, *Quantum Computation and Quantum Information*, Ch. 8: quantum noise and quantum operations.
2. Temme, Bravyi, Gambetta. Error mitigation for short-depth quantum circuits. *Phys. Rev. Lett.* 119, 180509 (2017).
3. Li and Benjamin. Efficient variational quantum simulator incorporating active error minimization. *Phys. Rev. X* 7, 021050 (2017).
4. Kim et al. Evidence for the utility of quantum computing before fault tolerance. *Nature* 618, 500–505 (2023).
5. Qiskit documentation:
<https://docs.quantum.ibm.com/guides/error-mitigation-and-suppression-techniques>